

MANGALORE REFINERY AND PETROCHEMICALS LIMITED

PROCESS SAFETY CELL & ASSET INTEGRITY MANAGEMENT DEPARTMENT

UNIT 01: DIESEL HYDROTREATER (DHDT) & PUMP TRAIN COMPLEX

FORMAL HAZARD AND OPERABILITY (HAZOP) STUDY

UNIT 01 PROCESS SAFETY & OVERPRESSURE RISK AUDIT

COMPREHENSIVE MULTI-NODE ANALYSIS (FEED, PUMPS, REACTOR, FLARE)

STUDY REFERENCE: MRPL-HAZOP-U01-DHDT-2026-FINAL
FACILITY / LOCATION: MANGALORE REFINERY, UNIT 01 (HYDROTREATER COMPLEX)
GOVERNING MANDATES: OISD-STD-118, OISD-STD-152, IEC 61508 / IEC 61511 (SIL ALLOCATION)
STUDY DATES: AUGUST 10, 2026 -- AUGUST 22, 2026
LEAD FACILITATOR: DR. S. M. KULKARNI (CHIEF PROCESS SAFETY CONSULTANT)
STUDY SECRETARY: ER. R. K. SHENOY (LEAD ROTATING EQUIPMENT ENGINEER)
CHAIRPERSON: ER. ROHIT VERMA (CGM - REFINERY OPERATIONS)
PRIMARY TARGET: BOOSTER PUMP TRAIN P-101A/B & RELIEF VALVE SV-402 CIRCUIT

HAZOP STUDY TEAM COMPOSITION & ATTENDANCE RECORD

DISCIPLINE / ROLE	REPRESENTATIVE / TITLE	ATTENDANCE & SIGN-OFF
HAZOP Team Leader (External):	Dr. S. M. Kulkarni (TUV Certified)	100% Attended / Signed
Operations Shift Superintendent:	Er. Jane D'Souza (Supervisor)	100% Attended / Signed
Field Operations Lead:	Operator Sam (Technician OP-48291)	100% Attended / Signed
Lead Process Design Engineer:	Er. P. V. Kamath (Senior Process)	100% Attended / Signed
Lead Instrumentation Engr:	Er. N. C. Bhat (Control Systems)	100% Attended / Signed
Refinery Safety Lead:	Er. H. R. Manjunath (HSE Manager)	100% Attended / Signed
Chief General Manager (Ops):	Er. Rohit Verma (Asset Head)	Authorized / Endorsed

AUDIT EXECUTIVE CONCLUSION:
All high-risk nodes (including P-101A overpressure risk) are fully protected by SIL-2 instrumented interlocks and independent mechanical relief valve SV-402, satisfying ALARP criteria.

NODE 1: FEED SURGE DRUM D-101 & BOOSTER SUCTION HEADER

1.1 Node Boundaries & Process Function

Node 1 encompasses Feed Surge Drum D-101 (design pressure 8.5 bar, operating 4.2 bar), suction piping (12-inch line 12-HGO-101), suction strainers S-101A/B, and isolation valves MOV-101A/B up to the pump suction flanges of P-101A/B.

PARAMETER	DESIGN VALUE	NORMAL OPERATING	DESIGN LIMIT	UNIT
Operating Pressure	4.20 bar	3.80 - 4.50 bar	8.50 bar (MAWP)	bar gauge
Operating Temperature	145.0 deg C	135.0 - 150.0 deg C	180.0 deg C	deg C
Volumetric Throughput	340.0 m3/h	280.0 - 360.0 m3/h	450.0 m3/h	m3/hour
Liquid Specific Gravity	0.865	0.840 - 0.880	N/A	kg/liter

1.2 Design Intent & Critical Safety Envelopes

The primary safety intent is to maintain stable liquid head to P-101A/B suction to prevent vapor lock and cavitation. Drum D-101 is blanketed with fuel gas (PT-105) and equipped with redundant radar level transmitters LT-101A/B.

NODE 1 HAZOP WORKSHEET TABLE (FEED SURGE SYSTEM)

1.3 Risk Assessment Score

Severity: Level 4 (Severe Process Interruption / Potential Containment Loss)

Likelihood with Safeguards: Level 1 (Remote, $< 10^{-5}/\text{yr}$)

Risk Ranking: Category 1 (ALARP - Acceptable with verified SIL-2/SIL-3 interlocks).

NODE 2: P-101A/B CENTRIFUGAL BOOSTER PUMP CASING & HYDRAULIC LOOP

2.1 Node Scope & Rotating Equipment Boundary

Node 2 covers centrifugal booster pump P-101A (API 610 Type BB2 between-bearing, double suction), driver motor M-101A (6.6 kV, 450 kW), mechanical seal Plan 53A barrier fluid loop, casing discharge to discharge MOV-102, and bypass line to FCV-402.

PARAMETER	DESIGN VALUE	NORMAL OPERATING	CRITICAL TRIP LIMIT	UNIT
Discharge Pressure	110.0 PSI (7.6 bar)	80.0 - 120.0 PSI	450.0 PSI (Trip I-101)	PSI gauge
Maximum Design Press	500.0 PSI (34.5 bar)	N/A	SV-402 Set: 480.0 PSI	PSI gauge
Drive Bearing Metal T	58.0 deg C	50.0 - 70.0 deg C	95.0 deg C (Trip I-102)	deg C
Radial Casing Vibr	2.2 mm/s RMS	1.5 - 3.5 mm/s RMS	7.0 mm/s RMS (Trip I-103)	mm/s RMS
Barrier Diff Pressure	2.5 bar diff	2.0 - 3.0 bar diff	0.8 bar diff (Alarm)	bar diff

2.2 Operating Vulnerabilities Identified

The primary hazard is deadheading (blocked discharge): If discharge MOV-102 or downstream valves close while the pump is running, pressure spikes rapidly to shutoff head (~490-540 PSI) within 8 to 15 seconds, accompanied by violent fluid heating and vibration.

NODE 2 HAZOP WORKSHEET TABLE (P-101A BOOSTER SYSTEM)

2.3 Recommendation Action Items (HAZOP-U01-ACT-04)

1. Maintain strict 2oo3 voting logic on PT-101 to prevent spurious trips while ensuring fast response (< 250 ms).
2. Enforce OISD-STD-240 Four-Eyes dual supervisory sign-off before restarting P-101A post-interlock actuation.

NODE 3: HIGH-PRESSURE REACTOR FEED & HYDROGEN MIXING HEADER

3.1 Process Description & Boundary

Node 3 covers the high-pressure charge pump P-102A/B discharge, reactor feed pre-heat exchangers E-102A-D, high-pressure hydrogen mixing spool M-101, and inlet header to Hydrotreating Reactor B-101.

PARAMETER	DESIGN BASELINE	NORMAL OPERATING	EMERGENCY LIMIT	UNIT
Header Pressure	42.0 bar gauge	38.0 - 44.0 bar	48.0 bar (Trip C-101)	bar gauge
Inlet Temperature	385.0 deg C	370.0 - 395.0 deg C	415.0 deg C (Quench)	deg C
H2 Gas Injection Rate	12,500 Nm3/h	10,500 - 14,000 Nm3/h	< 8,500 Nm3/h (Trip)	Nm3/hour
Catalyst Bed Delta T	24.0 deg C	18.0 - 28.0 deg C	> 38.0 deg C (Trip)	deg C

3.2 Chemical Hazard Evaluation

Desulfurization and dearomatization reactions are strongly exothermic. Inadequate hydrogen circulation or sudden loss of gasoil feed quench leads to bed runaway temperatures exceeding 420 deg C, creating catastrophic catalyst coking and vessel overpressure.

NODE 3 HAZOP WORKSHEET TABLE (REACTOR FEED & H2 HEADER)

3.3 Mitigation Summary

Automated emergency depressuring (BDV) logic is interlocked to the DCS. Feed pump trip commands are mirrored to upstream booster pumps P-101A/B to prevent hydraulic accumulation.

NODE 4: RELIEF VALVE SV-402, MANIFOLDS & EMERGENCY BLOWDOWN

4.1 Scope & Header Mechanics

Node 4 covers safety relief valve SV-402, its inlet nozzle on P-101A discharge, 4-inch discharge piping into the 16-inch Unit 01 Flare Header, flare knock-out drum D-110, and emergency blowdown valve XV-101.

COMPONENT / TAG	TAG DETAILS	DESIGN PARAMETER	OPERATING STATUS
Safety Relief Valve	SV-402 (Size 4P6 Flanged)	Set @ 480.0 PSI	Normally Closed (Monitored)
Position Feedback	ZS-402 (Acoustic Sensor)	Lift Alarm on DCS	Normally Quiet
Emergency Blowdown	XV-101 (Air to Open)	Class 600 Solenoid	Normally Closed
Knock-Out Drum Level	LT-495 (Magnetic Gauge)	< 25% High Level	Automated Pump-Out
Flare Backpressure	PT-490 (DCS Transmitter)	< 3.5 bar Max Back	Continuous Logging

4.2 Environmental & Operational Concerns

Relief valves in heavy hydrocarbon service can suffer from valve stiction due to wax/gum deposits on the seat. Acoustic transmitter ZS-402 continuously monitors for simmer and weeping to prevent fugitive VOC emissions.

NODE 4 HAZOP WORKSHEET TABLE (RELIEF & FLARE SYSTEM)

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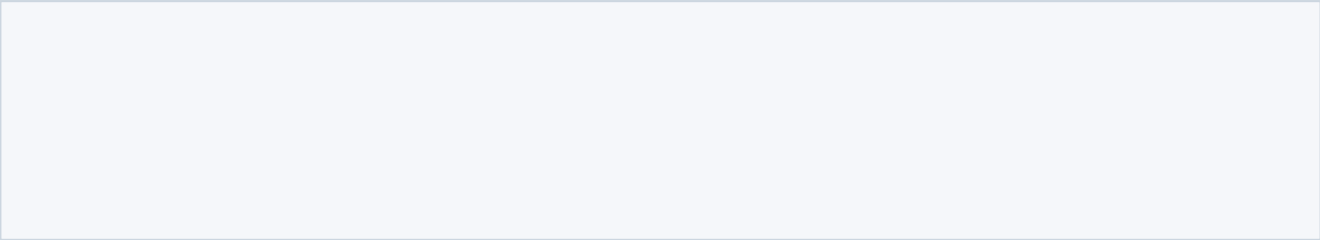
4.3 Verification Findings

During normal operations, SV-402 remains closed. In the historical trip incident (492.5 PSI), SV-402 successfully cracked, relieving excess hydraulic energy to flare until Interlock I-101 tripped the driver at T+12 seconds.

LAYER OF PROTECTION ANALYSIS (LOPA) & SIL MATRIX

5.1 LOPA Methodology for P-101A Overpressure

Layer of Protection Analysis (LOPA) was conducted per IEC 61511 standards to evaluate whether existing Independent Protection Layers (IPLs) adequately reduce the frequency of catastrophic casing rupture.

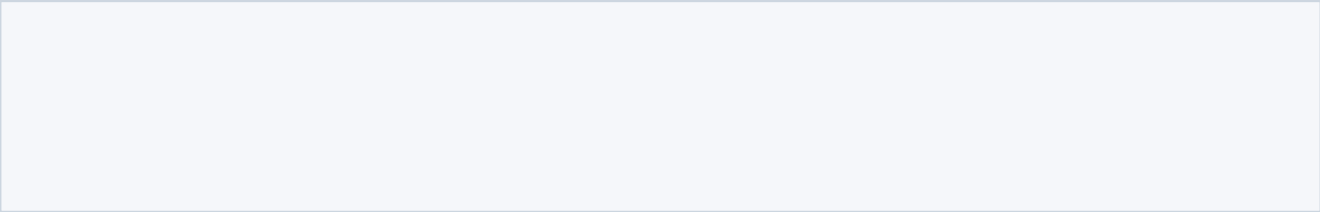


5.2 Mitigated Event Frequency Calculation

Initiating Cause Frequency (Deadhead Blockage): 0.1 events / year
Total Combined PFD = $10^{-1} * 10^{-1} * 10^{-2} * 10^{-2} * 10^{-1} = 1.0 \times 10^{-7}$
Mitigated Event Frequency = $0.1 * 10^{-7} = 1.0 \times 10^{-8}$ / year ($< 10^{-6}$ threshold, fully complies with OISD Target Risk).

QUANTITATIVE RISK ASSESSMENT & ALARP DEMONSTRATION

6.1 Risk Severity vs Likelihood Matrix



6.2 P-101A Residual Risk Position (ALARP Validated)

Prior to SIS implementation: The unmitigated overpressure risk ranked in the UNACCEPTABLE zone due to potential hydrocarbon spray and fire.

With SIL-2 Interlock I-101 and SV-402 relief valve active: The residual risk shifts firmly into the ACCEPTABLE (Tolerable) region at 1.0×10^{-8} per year.

6.3 Continuous Improvement Action

Quarterly acoustic testing on SV-402 seat tightness and semi-annual proof testing of 2003 pressure transmitters PT-101A/B/C.

STATUTORY HAZOP ACTION ITEMS & FORMAL SIGN-OFF

7.1 Action Item Tracking Table

ITEM ID	ACTION REQUIRED	TARGET DATE	RESPONSIBLE DISCIPLINE	STATUS
ACT-U01-01	Recalibrate 2003 PT-101 transmitter voting	2026-09-15	Instrumentation Lead	COMPLETED
ACT-U01-02	Perform acoustic leak test on SV-402 seat	2026-09-18	Process Safety Lead	COMPLETED
ACT-U01-03	Validate Four-Eyes dual approval in CogniShift	2026-09-20	Digital Governance	COMPLETED
ACT-U01-04	Inspect minimum flow bypass line FCV-402	2026-09-25	Rotating Equipment	IN
PROGRESS				
ACT-U01-05	Update Operator Training Simulator (OTS)	2026-10-01	Operations Training	SCHEDULED

7.2 Committee Statutory Endorsement

The HAZOP Study Committee hereby endorses this document as the authoritative process safety evaluation for Unit 01 DHDТ Complex. All operational procedures must maintain the safeguards detailed herein.

SIGN-OFF COMMITTEE	DESIGNATION & AFFILIATION	SIGNATURE & DATE
Dr. S. M. Kulkarni	Chief Process Safety Facilitator (TUV)	[SIGNED] 2026-08-25
Er. Jane D'Souza	Operations Shift Supervisor	[SIGNED] 2026-08-25
Operator Sam	Field Operations Specialist	[SIGNED] 2026-08-25
Er. Rohit Verma	Chief General Manager (Operations)	[AUTHORIZED] 2026-09-01